Ge Shi

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Education

University of California, Davis — *Oct. 2019 – Sept. 2024*
Ph.D. in Computer Science — *GPA: 3.8/4.0*

University of Massachusetts, Amherst — *Sept. 2017 – May. 2019*
M.S. in Computer Science, Machine Learning Track — *GPA: 3.9/4.0*

Zhejiang University — *Aug. 2013 – June. 2017*
B.S. in Control Science and Engineering — *GPA: 3.7/4.0*

Professional Experiences

Applied Scientist II — *Apr. 2025 – Present*
Amazon — *Seattle, WA*

- **SYNAPSE (core member scientist):** Amazon's centralized **agentic AI** system that automates 3,000+ long-horizon standard operating procedures (**SOPs**) for complex Seller, Buyer, and Vendor support cases that existing automation cannot handle. Serves a support operation of 1.1M cases per week, resolving cases at **89.5%** audited accuracy and about **half the cost** of a human-handled case.
- **Self-improving harness (module owner):** Lead the harness that lets SYNAPSE learn from its own production failures **without retraining**: it identifies issues, analyzes root causes, recommends fixes, re-verifies each fix against the original failure, and evaluates it before release. Running on 500+ seller cases per day, it fixed **20%** of the cases the agent previously failed.
- **Scalable knowledge compilation (co-inventor):** Turns large collections of operational documents into one organized, linked knowledge base for AI agents, with no per-domain setup. Compiled 8,000 documents into 18,000 pages, reused unchanged on six collections, and cut build time from 2–3 weeks to **under 4 hours**.
- **Compiling browser agents into scripts (lead inventor):** Turns one supervised run of an LLM browser agent into a reliable automation script that needs no model calls at run time, cutting inference cost to **zero**. Saves 20–50 seconds per 10-step task; the build cost pays for itself after about 9 runs.

Projects

Agentic AI — Procedural Memory, Continual Learning, Tool Use, Test-Time Scaling — *2026 – Present*

- **WASP-Mem (co-first author):** Investigated what matters in **procedural memory consolidation** for **continual-learning** agents. Introduced a training-free memory that distills execution traces and feedback into interlinked wiki pages, raising TheAgentCompany checkpoint pass rate by 4.6–6.1% over the strongest baseline with 57–61% less storage; one round of human diagnosis beat four rounds of self-judgment by 7.0–10.9%. arXiv 26
- **SLATE (co-first author):** Led the creation of a benchmark for **long-horizon**, **large-tool-space**, task-level tool selection, to study **test-time scaling** of AI agents. It pairs a 1,000-tool e-commerce library with a deterministic simulator and an automatic task-success metric that credits valid alternative trajectories; the best of six 3B–12B models completes only 5.0% of tasks. NeurIPS 26

Vision and Large Language Models — VLM, LLM, Pretraining, Finetuning, Alignment — *Aug. 2023 – Dec. 2024*

- Implemented a sub-**1B** parameters vision language model (VLM) through huggingface transformers following Llava's architecture with **SigLIP** and **QWen2.5-0.5B** which can be trained with a **single** T4 GPU on Google Colab.
- Performed **pretraining**, **visual instruction tuning**, and direct preference optimization (**DPO**) on the VLM model with the RLAIF-v dataset to achieve an accuracy of 61.9% on ScienceQA dataset.
- **Distributed** finetuned Llama-3 8B for instruction-following Chatbot on HealthCareMagic QA Dataset. Applied supervised fine-tuning techniques (**SFT**) such as full and parameter-efficient fine-tuning (**PEFT**) (**LoRA** and **qLoRA**) with **DeepSpeed** saving 99% parameters, 75% peak GPU memory and 55% training time.

Publications and Service

- **Publications** at NeurIPS and KDD spanning agentic AI, responsible AI, and AI for science. Google Scholar
- **Professional service:** long-term reviewer for NeurIPS, ICML, ICLR, AAAI, ACL Rolling Review, and IEEE VIS.

Technical Skills

Programming: Python, Java, C++, C, Matlab, HTML, CSS, Javascript, D3.js, Flask, R, SQL
Technologies: PyTorch, Huggingface, Scikit-learn, Keras, Pandas, Seaborn, NLTK, OpenCV, ROS, Git, AWS, Spark, Hadoop, Docker, Gradio, FastAPI, LangChain, MongoDB, FAISS, LATEX, MLflow, CI/CD, Lightning, WANDB, RAY
Skills: Agentic AI, LLM Agents, Machine/Deep Learning(ML/DL), Data Science(DS), Natural Language Processing(NLP), Computer Vision(CV)